



CHAPTER 5

Detailed Models of Overcontact Binary Stars



I can remember... sitting on a rooftop. Looking up and wondering what the stars were. I assumed I'd never know.

— *Hoid* —





AS PREVIOUSLY DISCUSSED, STARS undergoing mass transfer will generally experience RLOF during their lifetimes. In a binary where both stars are undergoing mass transfer, both stars may undergo RLOF simultaneously, resulting in a contact binary system in which both stars physically touch. Contact binaries are not uncommon – they form one in every approximately 100 low-mass main-sequence stars (Rucinski, 1998) and 39 in every 100 binary systems (Green et al., 2023).

Modelling their evolution generally offers understanding into difficult-to-probe stellar phenomena and evolutionary phases, such as the common envelope phase of binary evolution, as well as providing vital insights into the immediate pre-merger environment of a binary system. They are characterised by extremely short periods – typically with $P_{\text{orb}} \lesssim 2$ d, but potentially as low as $P_{\text{orb}} = 85$ min in theory (El-Badry et al., 2022) or ~ 167 min in practice (Koen, 2022). The lower bound of the period distribution is an element of some debate and ongoing research; Qian et al. (2020) finds it to be $0.15 \text{ d} \sim 200$ min, and Zhang et al. (2023) finds it to be at $0.12 \text{ d} \sim 170$ min. The contact binary phase of evolution can give rise to pre-supernova mergers, chemically abundant mass-transfer drastically altering the interior structure of the accretor (Chapter 3), and other modified evolutionary behaviour of the binary due to interaction effects stronger than that of wider-period binaries.

As a result of this close interaction, contact binaries may also be a formation channel for stellar exotica such as TŻOs, first proposed by Thorne & Żytkow (1977), where the capture of a neutron star by a supergiant companion in a close binary causes the formation of a TŻO (Taam et al., 1978). It is pertinent to note that these are not the *sole* formation channels for such exotica – for instance, a TŻO could be formed via a primary supernova kicking a nascent neutron star into its companion (Leonard et al., 1994; Podsiadlowski et al., 1995) or via collisions in clusters (Podsiadlowski et al., 1995) – but are instead potential formation channels.

Accurately modelling close-contact binaries, then, is of crucial importance to population synthesis models. Recent studies such as Menon et al. (2021b); Abdul-Masih et al. (2022); Sen et al. (2022) have explored some of the close-contact binaries in the Magellanic Clouds. Menon et al. (2021b) for example finds that for both Magellanic Clouds the majority of contact binaries are expected to have total masses of under $\sim 40 M_{\odot}$ and periods of less than ~ 1 d, and Abdul-Masih et al. (2022) finds that the sample of close-contact binaries that they analyse are likely to form with relatively longer periods than most systems.

Despite the importance of modelling close-contact binaries, only ~ 25 massive close-contact binaries are known to exist (Abdul-Masih, 2023). There are more known so-called W Ursae Majoris variables, with 700 compiled by Latković et al. (2021). These stars, typified by their namesake W Ursa Majoris, are low-mass binaries, typically with mass ratios of ~ 0.3 , and similar surface temperatures, though this study focuses on massive binaries. The paucity of observations is a reflection of the difficulty of distinguishing these systems from a more massive single star, especially since the two stars would share a common temperature (Kuiper, 1941; Latković et al., 2021) and thus similar observational properties. Spectroscopic observations also do not constrain physical parameters of contact binaries as well as they do for regular binaries (or single stars); for instance Gazeas et al. (2021) discuss the O’Connell effect and surface spots of the stars as being sources that modulate the luminosity of the binaries, but especially of the secondary.

Binary name	M_1	M_2	P	Z	$\log T_1$	$\log L_1$	$\log T_2$	Ref.
LSS 3074	14.8 ± 2.2	17.2 ± 4.2	2.183 ± 1	0.020^a	4.6 ± 0.2	5.15 ± 0.2	4.53 ± 0.5	A
MY Cam	37.7 ± 3.2	31.6 ± 4.2	$1.175^{+6}_{-1.175}$	0.020^a	4.62 ± 0.2	5.2 ± 0.5	4.59 ± 0.5	B
SMC 108086	16.9 ± 1.2	14.3 ± 5.1	0.883 ± 0.88	0.004^b	4.53 ± 0.1	4.608 ± 0.4	4.48 ± 0.5	C
TU Mus	16.7 ± 1.6	10.4 ± 1.4	1.387 ± 1	0.020^c	4.57 ± 0.2	5.02 ± 0.4	4.54 ± 0.5	D
V382 Cyg ^d	26.1 ± 3.2	19.0 ± 4.2	1.886 ± 1	0.020^e	4.57 ± 0.2	5.47 ± 0.4	4.58 ± 0.5	E
VFTS 352	28.6 ± 3.6	28.9 ± 1.8	$1.124^{+2}_{-1.124}$	0.010^f	4.63 ± 0.45	5.19 ± 0.5	4.61 ± 0.5	F

Table 5.1: Search parameters for the six close-contact binaries compiled by [Abdul-Masih et al. \(2022\)](#), as used in our fitting to the BPASS v2.2.1 grid. The uncertainties presented are not the physical uncertainties on their parameters, but instead represent the size of the range that we search within, e.g., $P = 1.175^{+6}_{-1.175}$ indicates that we match on all periods between 0 and 7.165 d inclusive. Masses and luminosities are measured relative to Solar, periods are measured in days, and temperatures measured in Kelvin. **Notes:** ^aSource paper assumes solar metallicity with no further qualification ^bUsing the mean metallicity of the SMC ^cBased on [Andersen & Grønbech \(1975\)](#) claiming 0.021 and [Terrell et al. \(2003\)](#) confirming a comparison with synthetic spectra “does not support a carbon content, and therefore metallicity, far from the solar value” ^dA mass ratio inversion occurred for this binary (see [Martins et al., 2017](#)) and hence we use the definition of “primary” from within the paper when reporting in this table. ^eUsing the mean metallicity of the Milky Way ^fAs [Crowther et al. \(2018\)](#) reports “half-solar” for the Tarantula Nebula, the host nebula of VFTS 352. **References:** A: [Raucq et al. \(2017\)](#), B: [Lorenzo et al. \(2014\)](#), C: [Hilditch et al. \(2005\)](#), D: [Penny et al. \(2008\)](#), E: [Martins et al. \(2017\)](#), F: [Almeida et al. \(2015\)](#)

5.1 | Observations of contact binaries

There’s a place called Wonderland.

— FIFTH OF THE SIX IMPOSSIBLE THINGS

Observations of contact binaries date back to the 1940s ([Kuiper, 1941](#)), and in the past eighty years the catalogue has grown to somewhere between $\sim 42\,000$ ([Petrosky et al., 2021](#)) and $\sim 86\,000$ ([Qian et al., 2020](#)) contact binaries, representing between 26.7 and 54.8 per cent of all observed binaries (those from the Washington Double Star catalogue; [Mason et al. 2001](#)). The binaries we study in this chapter are summarised in Table 5.1, and were originally compiled by [Abdul-Masih et al. \(2022\)](#). Their sample focused on measuring the period stability of contact binaries and is thus a subset of the known sample. They rejected a handful of other non O+O overcontact binaries, for reasons that include short-period von Zeipel-Kozai-Lidov oscillations, incomplete photometric / radial velocity fits, non-MS binaries, unknown separations, and irregular light curves.

Ages for the six binaries have not been published, though a soft restriction can be placed on their evolutionary status; since [Abdul-Masih et al. \(2022\)](#) compile specifically massive MS contact binaries, their interactions must take place whilst the progenitor stars are on their respective main sequences.

5.1.1 | Progenitor history

We now discuss each of the overcontact systems in detail, providing observational context to each system. For each binary, we list a selection of key measurements, including the fillout factor for each star. A fillout

factor is a measurement of the degree to which a system overfills its Roche lobes, with $1 < f < 2$ denoting an overcontact binary, and a high fillout factor representing a more swelled and in-contact system. This definition of the fillout factor is a property of the *system*, not the *constituent stars*, and originates from [Mochnicki & Doughty \(1972\)](#).

LSS 3074

The binarity of LSS 3074 ([Raucq et al., 2017](#)) was first observed in 1985 by [Morrell & Niemela \(1990\)](#) with an observed period of 2.19 d, with its HeI and HeII spectra causing the authors to conclude that it was an O4f+ and O6-7 binary, with inclination-weighted masses of $M_1 \sin^3 i = 8 M_\odot$ and $M_2 \sin^3 i = 9 M_\odot$. With later measurements of the inclinations to be $i \approx 24^\circ$, these masses were refined to $M_1 = 17.2 M_\odot$ and $M_2 = 14.8 M_\odot$. It is located behind the Coalsack Nebula, at a distance of 2256 pc, with a B-band magnitude of 12.81 mag ([Stephenson & Sanduleak, 1971](#)). It has a fillout factor $f = 1.05$.

MY Camelopardalis

MY Camelopardalis (MY Cam; [Lorenzo et al., 2014](#)) is a binary in the cluster Alicante 1, and its binarity was established in 2003 by [Negueruela & Marco \(2003\)](#). It is likely to be a very massive merger progenitor, though is barely in contact with $f = 1.01$ ([Abdul-Masih et al., 2022](#)).

SMC 108086

OGLE SMC-SC10 108086 (SMC 108086; [Hilditch et al., 2005](#)) is a O9 + O9.5/B0 binary. It is a system deeply in contact, with $f = 1.70$ ([Hilditch et al., 2005](#); [Abdul-Masih et al., 2022](#)) – the deepest known – and has a short period of $P = 0.88$ d. It is located at a distance of 56.5 kpc, in the Small Magellanic Cloud (SMC).

TU Muscae

TU Muscae (TU Mus; [Penny et al., 2008](#)) is an O8V + B0V binary discovered by [Oosterhoff \(1928, 1930\)](#). Its binarity was confirmed as late as 1971, by [Knipe \(1971\)](#). The system was profiled by [Andersen & Grønbech \(1975\)](#), who concluded that, based on comparison to theoretical models by [Stothers \(1972, 1973\)](#), the system was of solar composition ($Z = 0.021 \approx Z_\odot$) and on the main sequence with $X_H = 0.739$. It has a fillout factor of $f = 1.12$.

V382 Cygni

V382 Cygni (V382 Cyg; [Martins et al., 2017](#)) is an O6.5 + O6 binary whose components are oblate spheroids (deduced from the β -Lyrae light curves; [Pearce 1952](#)). It was originally discovered by [Morgenroth \(1935\)](#); [Petrov \(1946\)](#). It has a fillout factor of $f = 1.10$.

VFTS 352

VFTS 352 was detected in the VLT-FLAMES survey of the Tarantula Nebula (Evans et al., 2011) and is the most recent detection in our sample. The Tarantula Nebula / 30 Doradus is a large H(II) region in the Large Magellanic Cloud (LMC). The progenitor of VFTS 352 is likely an equal mass, short-period binary with no strong helium or nitrogen surface abundances (Abdul-Masih et al., 2019), which may be due to biases in the CNO abundances, or similar, in that atmospheric study. It has a fillout factor of $f = 1.28$.

5.1.2 | Data matching with BPASS v2.2

To constrain the nature and prior evolution of these contact binaries we attempt to match stellar models from the BPASS v2.2 models to these binaries. We perform our data matching using the HOKI code (Stevance et al., 2020). We look for matches in masses, periods, temperature / luminosity of the primary, and temperature of the secondary. The luminosity of the secondary in each case is less constrained, and in some cases unpublished. The search parameters are shown in Table 5.1. We query the BPASS v2.2.1 dataset and record whether or not they enter the (widened) parameter values. Parameters are widened as some of the periods are tighter than the BPASS bins and thus can be missed in the search.

We note three key caveats in our search criteria.

1. We have performed our matching based on parameters which are difficult to constrain for an eclipsing binary. The better choice would be to match primarily on the masses / radii, which are known more precisely for eclipsing binaries (as the measurement dependence is on the times of first and second contact, and the relative velocity of the stars, which can be determined through radial velocity measurements). The rationale is that the BPASS v2.2 models use the Hurley et al. (2000) approximations, which become highly approximate during the mass transfer phase – resulting in the mass, temperature, and luminosity of the companion becoming more approximate during mass transfer. Consequently, we must preference matching on the donor parameters over the accretor parameters, though we include the accretor mass and effective temperature to allow for it to have some influence on the output models. This also means that we must be content with the wider error bars on the search parameters this necessitates. Matching on six parameters instead of four also means that our results are more likely to be outliers; they may either be overfits or not fit at all, with both a result of necessarily matching on more parameters.
2. As a consequence of the first caveat, the search error bars are widened, in some cases substantially. The known parameters by Abdul-Masih et al. (2022) are so tightly constrained that the parameter grid spacing in BPASS becomes significant, and thus the search bars must be widened to account for this. This is also made manifest by the approximations in BPASS v2. BPASS uses the approximations of Hurley et al. (2000) for the secondary radius, temperature, and luminosity. Thus, we widen our parameters for the secondary to compensate for this approximation.
3. As alluded to previously, luminosities are not uniformly reported for the companions across the grid.

In particular, the companion luminosity for VFTS 352 is unreported, and thus we approximate it using the Stefan-Boltzmann law,

$$\frac{L}{L_{\odot}} = \left(\frac{R}{R_{\odot}} \right)^2 \left(\frac{T}{T_{\odot}} \right)^4, \quad (5.1)$$

where $T_{\odot} = 5778$ K. This induces two assumptions: that the companion is a perfect blackbody, and that it is spherical. The first assumption is typically assumed regardless, and the secondary is a necessary condition of our 1D stellar evolution approach. Although the star is, physically, distorted with a fillout factor of $f = 1.28$, the key assumption underpinning 1D stellar evolution is spherical symmetry.

The matching values HOKI finds are shown in Table 5.2. The primary models are also shown on Figure 5.1. We also report in the table the inferred ages at match, which lie between $\mu_{\log(\text{age})} = 6.13$ (for MY Cam) and $\mu_{\log(\text{age})} = 7.14$ (for TU Mus).

To determine the likely future history of this sample, we must use the TUI code to match pre- and post-supernova models. To that end, in Table 5.3, we enumerate every resultant match to the observed parameters from Table 5.1. For each match, we display the masses, radii, and effective temperatures of each star in the binary. We also track whether the stars are – at the time of match – filling their Roche lobes.

Our modelling shows that all of these contact binaries survive their contact phase, eventually exiting to continue on as semi-detached and detached binaries. Importantly, none overflow their $\mathcal{L}_2$ points – which is commonly taken as a precursor signal of a merger event in other codes such as POSYDON (Fragos et al., 2023) – and none of them merge. As they evolve, as shown in Table E.1, the primary typically ends up as a helium star or WR star, and the secondaries become more massive, typically with a post-contact mass ratio inversion occurring.

The full future history is completed by examining the remnants both after the primary and secondary supernova. To account for the uncertainties present in the natal kick prescription (Chapter 4), we consider three kick prescriptions: the Hobbs et al. (2005) and Richards et al. (2023) kicks from Chapter 4, and the bimodal Verbunt et al. (2017) kick. The Hobbs et al. (2005) and Richards et al. (2023) kicks were described in Chapter 4, and the Verbunt et al. (2017) kick motivates its choice with a second peak at lower kick velocities to account for phenomena such as electron-capture supernovae and USSNe. It samples the kick velocity from a modified Maxwellian distribution, a la Hobbs et al. (2005), though the distribution is given by

$$P(v_k) dv_k = \sqrt{\frac{2}{\pi}} v_k^2 \left[\frac{w}{\sigma_1^3} \exp\left(-\frac{v_k^2}{2\sigma_1^2}\right) + \frac{1-w}{\sigma_1^3} \exp\left(-\frac{v_k^2}{2\sigma_2^2}\right) \right], \quad (5.2)$$

where $\sigma_1 = 75 \text{ km s}^{-1}$ and $\sigma_2 = 316 \text{ km s}^{-1}$ are the distribution parameters and $w = 0.42$ is the weight of the lower peak. For each binary, we compute the post-supernova kick parameters M_1, M_2, a_0 and e_0 in the same way as in Chapter 4, and – after the secondary supernova – use these to calculate the coalescence time in Gyr.

For each system, we record the type of binary left after the secondary supernova – NS-BH, Black Hole – White Dwarf Binary (BH-WD), Neutron Star – White Dwarf Binary (NS-WD), BNS, Binary Black Hole

Binary	BPASS v2.2.1 matches	$M_{1,ZAMS}$	$M_{2,ZAMS}$	q	P_{ZAMS}	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$
LSS 3074	sneplot-z020-30-0.3-0.6	30.0	9.0	0.3	3.98	6.54	0.64
	sneplot-z020-25-0.4-0.6	25.0	10.0	0.4	3.98	6.80	0.12
	sneplot-z020-25-0.5-0.6	25.0	12.5	0.5	3.98	6.80	0.13
	sneplot-z020-24-0.4-0.6	24.0	9.6	0.4	3.98	6.83	0.09
	sneplot-z020-24-0.5-0.6	24.0	12.0	0.5	3.98	6.83	0.09
MY Cam	sneplot-z020-50-0.5-0.6	50.0	25.0	0.5	3.98	6.35	0.67
	sneplot-z020-50-0.4-0.8	50.0	20.0	0.4	6.31	6.31	0.72
	sneplot-z020-50-0.5-0.8	50.0	25.0	0.5	6.31	6.21	0.81
	sneplot-z020-50-0.6-0.8	50.0	30.0	0.6	6.31	6.13	0.88
	sneplot-z020-40-0.7-0.8	40.0	28.0	0.7	6.31	6.43	0.66
SMC 108086	sneplot-z004-24-0.5-0.4	24.0	12.0	0.5	2.51	6.47	0.77
	sneplot-z004-23-0.4-0.4	23.0	9.2	0.4	2.51	6.63	0.63
	sneplot-z004-23-0.5-0.4	23.0	11.5	0.5	2.51	6.62	0.65
	sneplot-z004-22-0.4-0.4	22.0	8.8	0.4	2.51	6.70	0.57
	sneplot-z004-22-0.5-0.4	22.0	11.0	0.5	2.51	6.70	0.57
	sneplot-z004-21-0.4-0.4	21.0	8.4	0.4	2.51	6.77	0.50
	sneplot-z004-21-0.5-0.4	21.0	10.5	0.5	2.51	6.76	0.51
	sneplot-z004-20-0.4-0.4	20.0	8.0	0.4	2.51	6.79	0.49
	sneplot-z004-20-0.5-0.4	20.0	10.0	0.5	2.51	6.78	0.49
	sneplot-z004-19-0.4-0.4	19.0	7.6	0.4	2.51	6.80	0.49
TU Mus	sneplot-z020-25-0.3-0.4	25.0	7.5	0.3	2.51	6.69	0.56
	sneplot-z020-24-0.3-0.4	24.0	7.2	0.3	2.51	6.71	0.56
	sneplot-z020-23-0.3-0.4	23.0	6.9	0.3	2.51	6.77	0.52
	sneplot-z020-16-0.6-0.4	16.0	9.6	0.6	2.51	7.14	0.05
V382 Cyg	sneplot-z020-50-0.2-0.6	50.0	10.0	0.2	3.98	6.44	0.56
	sneplot-z020-40-0.3-0.6	40.0	12.0	0.3	3.98	6.52	0.54
	sneplot-z020-40-0.4-0.6	40.0	16.0	0.4	3.98	6.51	0.55
	sneplot-z020-35-0.4-0.6	35.0	14.0	0.4	3.98	6.55	0.56
	sneplot-z020-30-0.4-0.6	30.0	12.0	0.4	3.98	6.60	0.57
	sneplot-z020-30-0.5-0.6	30.0	15.0	0.5	3.98	6.59	0.57
VFTS 352	sneplot-z010-40-0.5-0.6	40.0	20.0	0.5	3.98	6.55	0.53

Table 5.2: Matches from the HOKI (Stevance et al., 2020) software. Masses are measured in solar masses, periods in days, and ages in years. Ages were computed as the mean of the ages in log-space where a match occurs, and similar for the standard deviations.

ZAMS parameters												Match Parameters			
System Name	$M_1 / M_\odot$	$M_2 / M_\odot$	$\log P / \text{d}$	$M_1 / M_\odot$	$M_2 / M_\odot$	$\log R_1 / R_\odot$	$\log R_2 / R_\odot$	$\log T_1 / \text{K}$	$\log T_1 / \text{K}$	Star 1 RLOF	Star 2 RLOF				
LSS 3074	55.00	15.00	0.50	36.02	17.92	1.03	0.81	4.54	4.58	✓	✓				
LSS 3074	50.00	15.00	0.90	26.98	22.14	1.06	0.81	4.53	4.61	✓	✓				
TU Mus	30.00	5.00	0.70	15.46	8.58	0.91	0.80	4.55	4.52	✓	✓				
TU Mus	55.00	10.00	0.90	26.25	17.99	0.96	0.66	4.55	4.56	✓	✓				
V382 Cyg	45.00	15.00	0.50	27.98	19.29	0.98	0.90	4.54	4.61	✓	✓				
V382 Cyg	50.00	15.00	0.50	31.74	18.66	1.00	0.86	4.54	4.61	✓	✓				
V382 Cyg	50.00	20.00	0.50	36.95	23.71	1.07	0.90	4.55	4.58	✓	✓				
V382 Cyg	45.00	25.00	0.50	28.55	29.34	0.97	0.93	4.54	4.56	✓	✓				
V382 Cyg	35.00	10.00	0.60	20.64	14.32	0.99	0.92	4.54	4.56	✓	✓				
V382 Cyg	40.00	10.00	0.60	23.91	13.98	0.99	0.88	4.57	4.57	✓	✓				
V382 Cyg	55.00	10.00	0.60	30.19	13.38	0.97	0.80	4.56	4.61	✓	✓				
V382 Cyg	40.00	15.00	0.60	25.57	20.74	1.03	0.96	4.54	4.60	✓	✓				
V382 Cyg	45.00	15.00	0.60	30.64	19.51	1.07	0.94	4.54	4.58	✓	✓				
V382 Cyg	45.00	20.00	0.60	27.08	26.03	0.99	0.88	4.54	4.61	✓	✓				
V382 Cyg	45.00	15.00	0.70	26.15	20.82	1.01	0.83	4.54	4.61	✓	✓				
V382 Cyg	50.00	20.00	0.70	27.80	29.75	0.98	0.85	4.58	4.62	✓	✓				
LSS 3074	30.00	15.00	0.60	17.23	22.90	1.01	0.84	4.53	4.59	✓	✗				
TU Mus	30.00	10.00	0.50	16.44	14.94	0.90	0.73	4.56	4.56	✓	✗				
TU Mus	30.00	10.00	0.60	15.39	16.34	0.93	0.67	4.56	4.54	✓	✗				
TU Mus	30.00	5.00	0.80	14.81	9.36	0.96	0.71	4.55	4.55	✓	✗				
V382 Cyg	35.00	15.00	0.50	19.62	21.78	0.92	0.82	4.57	4.61	✓	✗				
V382 Cyg	35.00	20.00	0.50	20.82	28.25	0.96	0.90	4.56	4.61	✓	✗				
V382 Cyg	40.00	20.00	0.50	16.71	28.23	0.91	0.89	4.59	4.61	✓	✗				
V382 Cyg	35.00	20.00	0.60	20.84	30.86	1.02	0.89	4.56	4.61	✓	✗				
V382 Cyg	35.00	10.00	0.70	20.12	15.43	1.03	0.91	4.54	4.57	✓	✗				

Table 5.3: Matches from the TU1 code. All parameters are rounded to two decimal places, and are given as at the first moment of matching. Only binaries that have at least one component filling their Roche lobe are shown in the list, with the balance listed in Table D.1.

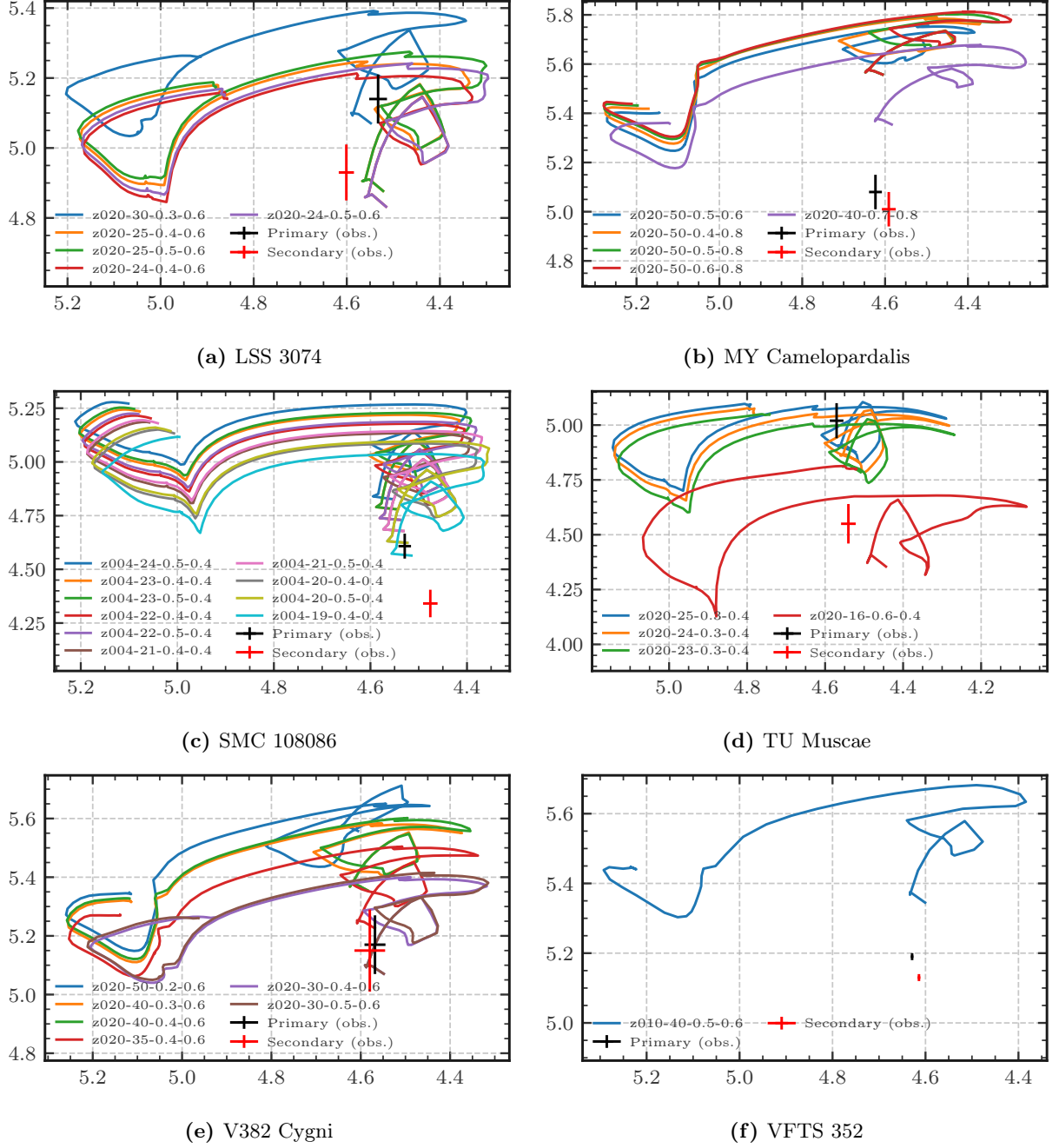
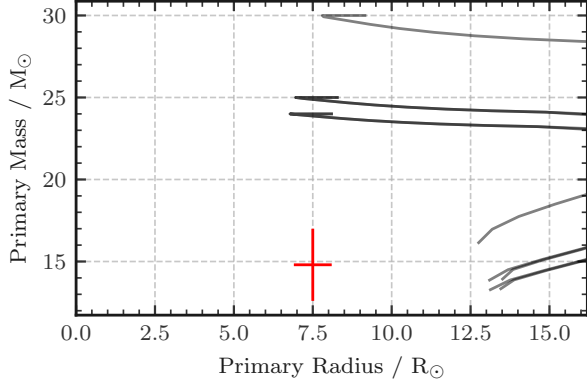
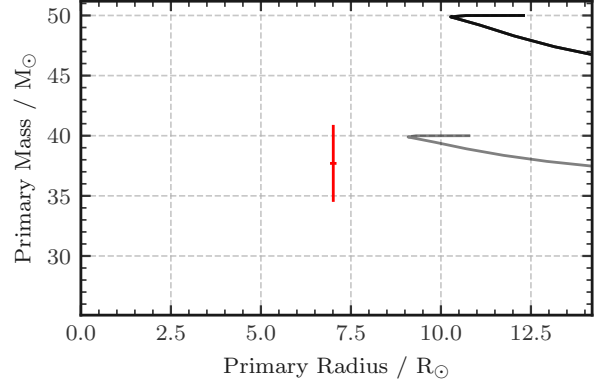


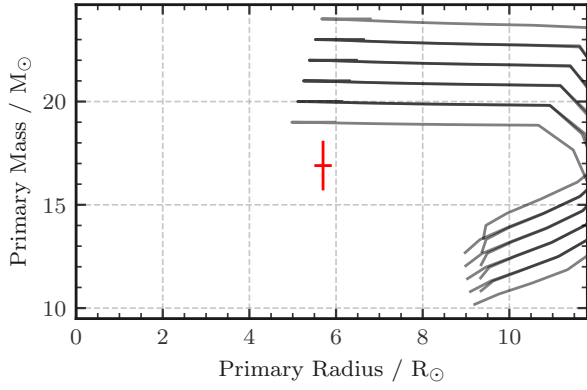
Figure 5.1: BPASS v2.2.1 primary models for the matches obtained by HOKI for the parameters in Table 5.1. The legend key is the BPASS stellar model file name (omitting the “sneplot-” prefix common to every entry), and is broken down into its parts in Table 5.2. The overplotted observed systems do not match every system, and the search error bars were widened somewhat as a result (see Table 5.1 for details).



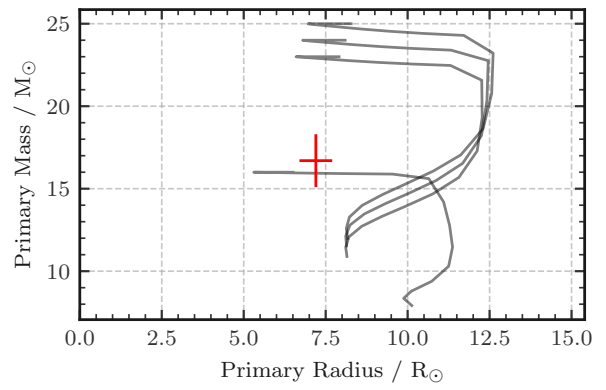
(a) LSS 3074



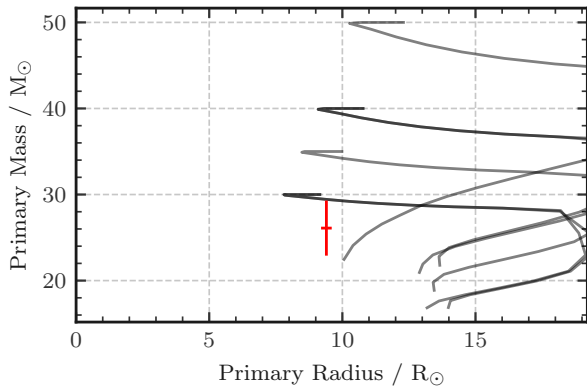
(b) MY Camelopardalis



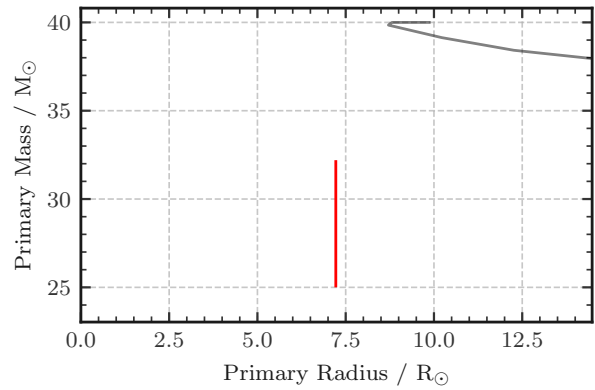
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure 5.2: BPASS v2 model tracks for our six close-contact binaries. Models shown are restricted to only during their core hydrogen burning phase, which for this plot we assume to be when the helium core mass $M_{\text{He}} < 10^{-5} M_{\odot}$. The x -axis is limited to double the maximum value of the observed radius.

(BBH). If the mean eccentricity is less than zero or strictly greater than one, we record it as “Unbound”, and present the percentage of systems in which the binary unbinds. If the binary remains bound, we present the coalescence time, which is allowed to be arbitrarily large (with the largest being 1.75 Pyr). In practice, arbitrarily large coalescence times are likely unphysical, with Pyr coalescence times approaching the end of the stelliferous era. Nevertheless, we include them as predictions made by the integration of the equations of [Peters \(1964\)](#).

We assume that a white dwarf is left if the CO core mass of the star is less than $1.38 M_{\odot}$, the total mass of the star is less than $1.5 M_{\odot}$, or if core carbon burning does not occur. Otherwise, we assume that remnants with $M_{\text{rem}} \leq 2.5 M_{\odot}$ leave behind a NS, and otherwise a BH is left.

Across the three kick models, in approximately 55.2 per cent of the binaries, the binary remains bound after the secondary supernova. Most of the unbound models (44.4 per cent) originate from the [Verbunt et al. \(2017\)](#) kick, and the [Richards et al. \(2023\)](#) kick from Chapter 4 contributes 17.8 per cent of the unbound models. The bound models typically have initial post-secondary-supernova periods of tens of years, with a notable outlier in MY Camelopardalis, which has two systems with separations $> 100 R_{\odot}$, creating large periods.

Binary	$\bar{a}_0 / R_{\odot}$	$\bar{e}_0[no.dim]$	$\bar{P}_0 / \text{yr}$
LSS 3074	6.77 ± 0.10	0.55 ± 0.09	1.433 ± 0.034
MY Cam	70.17 ± 0.10	0.68 ± 0.07	56.23 ± 0.05
SMC 108086	8.05 ± 0.16	0.67 ± 0.12	1.72 ± 0.05
TU Mus	7.69 ± 0.20	0.59 ± 0.12	2.04 ± 0.05
V382 Cyg	17.38 ± 0.08	0.66 ± 0.06	5.376 ± 0.034
VFTS 352	29.62 ± 0	0.9 ± 0	7.31 ± 0

Table 5.4: The mean post-secondary-supernova parameters for the bound remnants of our binary models. Uncertainties are the 1σ uncertainties from taking the mean across all parameters, and the uncertainty on VFTS 352 is due to there only being one bound remnant in the calculation. Individual kick prescriptions are not distinguished for this calculation.

A wide variety of remnants are possible; these are summarised below:

1. LSS 3074

- NS-NS (9 occurrences)
- NS-WD (3 occurrences)

2. MY Cam

- NS-NS (5 occurrences)
- BH-NS (4 occurrences)
- NS-WD (3 occurrences)
- NS-BH (1 occurrence)

3. SMC 108086

- NS-NS (15 occurrences)
- NS-WD (2 occurrences)

4. TU Mus

- NS-NS (7 occurrences)
- NS-WD (2 occurrences)

5. V382 Cyg

- NS-NS (7 occurrences)
- NS-WD (1 occurrence)

6. VFTS 352

- BH-BH (1 occurrence)

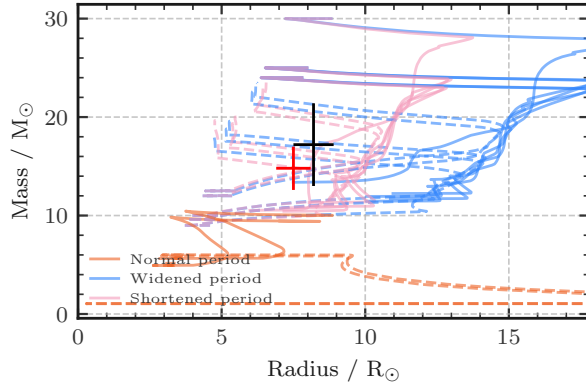
We present the full table of results in Appendix E. We can see a number of extremely low-mass WDs, the lightest of which are $0.28 M_{\odot}$, barely above the current lowest-mass WD ($0.17 M_{\odot}$; [Kilic et al., 2007](#)). It is possible that these may be numerical artefacts, and that the supernova has either destroyed the star leaving effectively no remnant, or that the stars merged during the supernova event. This would be compounded by the stars appearing to be unbound, with high frequency. The TUI code does not record whether a merger occurs during the supernova, so this is not possible to test given the current architecture of the data.

5.2 | Creation of bespoke binary star evolution models

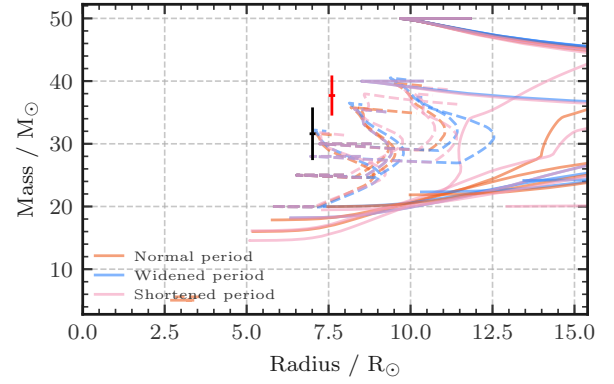
Having now established the ZAMS parameters for our binaries, we can take these as input to our STARS code to determine their future history. For each matched model, we use the initial masses and periods as inputs to the STARS evolution code. We use the AOTEAROA STARS binary star evolution program as described in Chapters 2 and 3.

While we use the periods from the BPASS model fits we also expand our search to two extra periods per pair of initial masses. The first is the fiducial period, as reported from the HOKI code. The second is a set of binaries with identical parameters except for periods 0.1 dex higher, and the third is binaries with periods 0.1 dex lower. Our aim, in evaluating the expanded period grid, is to determine the sensitivity of the grid to this parameter choice, given that the major aim of this chapter is to investigate, effectively, short-period binaries.

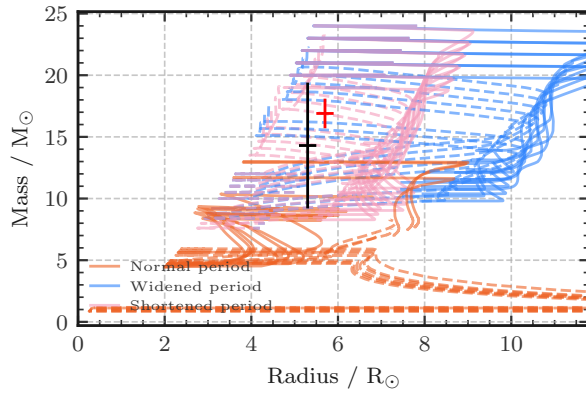
The results of our forward evolution are shown in Appendix D. We also show summary plots for the six close-contact binaries in Figure 5.3, where we plot the primary mass against the primary radius. As the six contact binaries are all eclipsing binaries, the masses and radii are the most well-constrained parameters.



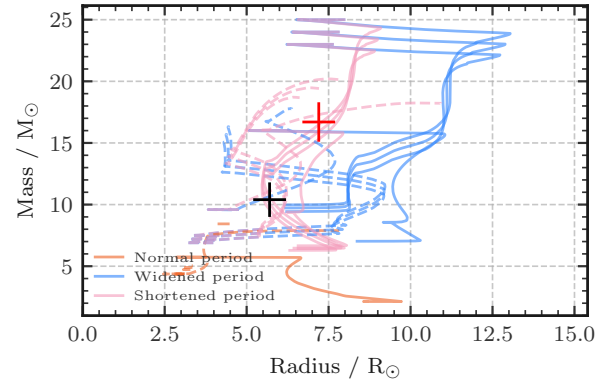
(a) LSS 3074



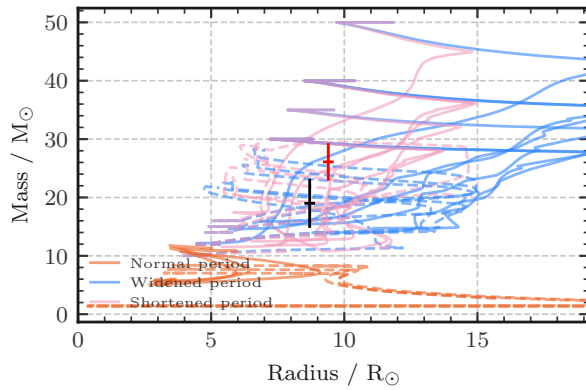
(b) MY Camelopardalis



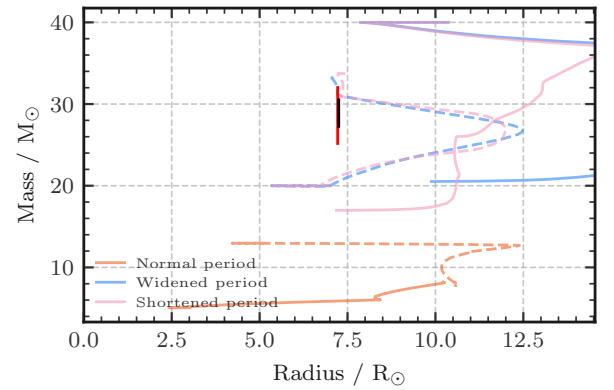
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure 5.3: Detailed model tracks for our six close-contact binaries. Solid (dashed) lines represent the donor (accretor), which has its observed parameters shown in red (black). Models shown are restricted to only during the core hydrogen burning phase of the donor. The x -axis is limited to double the maximum value of the observed radius of either star.

As a result, however, there is a dearth of matching models. LSS 3074, TU Muscae, and V382 Cygni are the only three systems with any matches to the observed masses and periods, which are enumerated in Table 5.5. We also take each of the matching models and display them in Figure 5.4. There are 14 matching models in total.

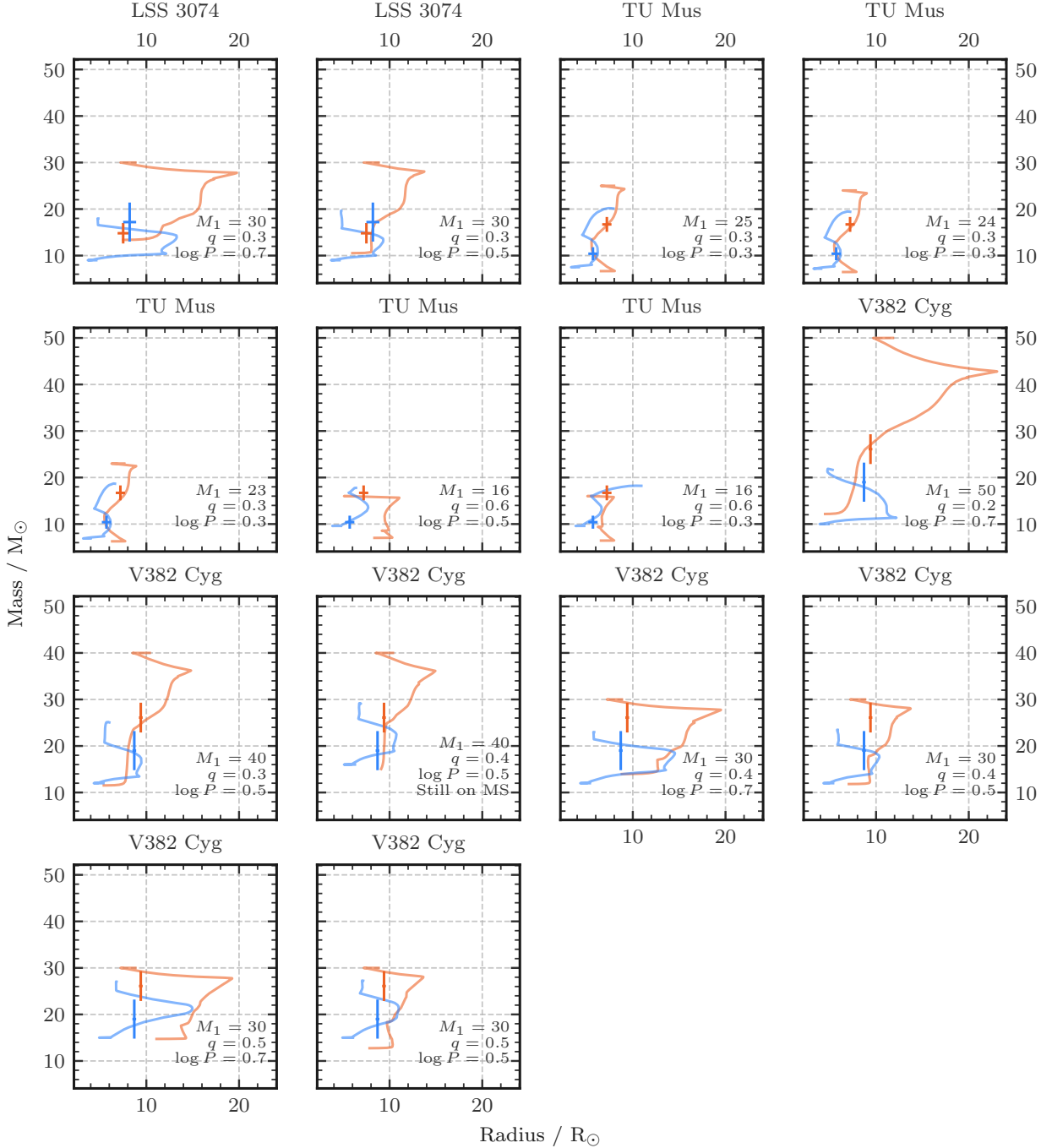


Figure 5.4: The mass-radius plots for our donor (orange) and accretor (blue) models that match their observations. The tracks are restricted to whilst the donor is still undergoing core hydrogen fusion, with the exception of one model which terminated on the main sequence and is marked on its plot.

Binary	ZAMS parameters				Ages			Match details			
	Z	$M_1 / M_\odot$	q	$\log P / d$	Begin / Myr	End / Myr	Length / kyr	$M_1 / M_\odot$	$R_1 / R_\odot$	$M_2 / M_\odot$	$R_2 / R_\odot$
LSS 3074	z020	30	0.3	0.7	6.1087	6.1141	5.43	13.37 $\pm$ 0.01	7.40 $\pm$ 0.33	17.96 $\pm$ 0.00	4.82 $\pm$ 0.00
LSS 3074	z020	30	0.3	0.5	5.4656	6.1309	665.33	13.73 $\pm$ 0.63	8.03 $\pm$ 0.03	18.39 $\pm$ 0.27	4.82 $\pm$ 0.03
TU Mus	z020	25	0.3	0.3	3.2888	3.2934	4.63	16.50 $\pm$ 0.59	7.20 $\pm$ 0.29	11.55 $\pm$ 0.54	6.17 $\pm$ 0.06
TU Mus	z020	24	0.3	0.3	3.5528	3.5574	4.61	16.01 $\pm$ 0.57	7.24 $\pm$ 0.27	10.92 $\pm$ 0.50	6.15 $\pm$ 0.04
TU Mus	z020	23	0.3	0.3	3.8383	3.8415	3.28	15.77 $\pm$ 0.41	7.42 $\pm$ 0.17	10.07 $\pm$ 0.32	6.13 $\pm$ 0.01
TU Mus	z020	16	0.6	0.5	5.3727	7.1860	1813.26	15.91 $\pm$ 0.01	7.18 $\pm$ 0.29	9.58 $\pm$ 0.00	4.36 $\pm$ 0.04
TU Mus*	z020	16	0.6	0.3	5.1722	6.9546	1782.43	15.90 $\pm$ 0.01	7.18 $\pm$ 0.29	9.58 $\pm$ 0.00	4.37 $\pm$ 0.04
TU Mus*	z020	16	0.6	0.3	7.3024	7.3026	0.27	15.12 $\pm$ 0.01	7.69 $\pm$ 0.01	10.33 $\pm$ 0.01	5.02 $\pm$ 0.01
V382 Cyg	z020	50	0.2	0.7	3.0350	3.0399	4.96	27.18 $\pm$ 0.19	9.43 $\pm$ 0.11	21.16 $\pm$ 0.17	4.79 $\pm$ 0.04
V382 Cyg	z020	40	0.3	0.5	2.9967	2.9982	1.45	24.70 $\pm$ 0.19	9.40 $\pm$ 0.11	20.05 $\pm$ 0.19	6.40 $\pm$ 0.30
V382 Cyg	z020	40	0.4	0.5	3.2253	3.6514	426.16	23.61 $\pm$ 0.41	9.54 $\pm$ 0.03	26.69 $\pm$ 0.21	6.72 $\pm$ 0.02
V382 Cyg	z020	30	0.4	0.7	2.3847	2.6565	271.78	29.24 $\pm$ 0.03	9.43 $\pm$ 0.10	12.00 $\pm$ 0.00	4.70 $\pm$ 0.01
V382 Cyg	z020	30	0.4	0.5	2.2749	2.5998	324.97	29.17 $\pm$ 0.04	9.40 $\pm$ 0.12	12.00 $\pm$ 0.00	4.70 $\pm$ 0.01
V382 Cyg	z020	30	0.5	0.7	2.3662	2.6543	288.04	29.24 $\pm$ 0.03	9.42 $\pm$ 0.10	14.98 $\pm$ 0.00	5.48 $\pm$ 0.02
V382 Cyg	z020	30	0.5	0.5	2.2768	2.6024	325.57	29.17 $\pm$ 0.04	9.40 $\pm$ 0.12	14.98 $\pm$ 0.00	5.49 $\pm$ 0.02

Table 5.5: The matches in primary mass / primary radius to our overcontact models. Shown here are the ZAMS parameters. An asterisk next to the star name indicates that the binary undergoes two periods where it matches the donor mass / radius, and each match is shown. Quoted uncertainties are the 1σ confidence levels, and all quantities are (for brevity) always rounded to 2dp.

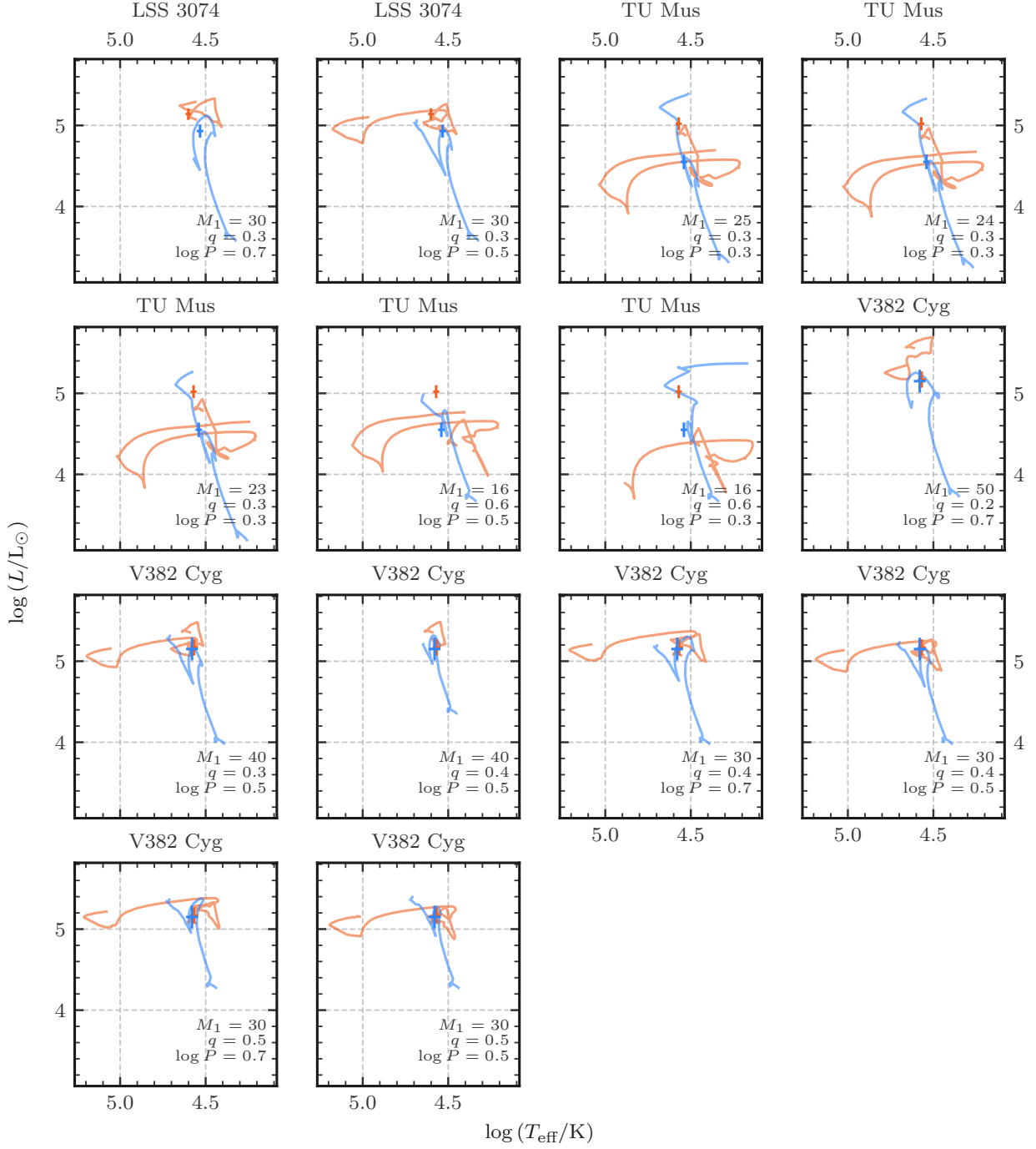
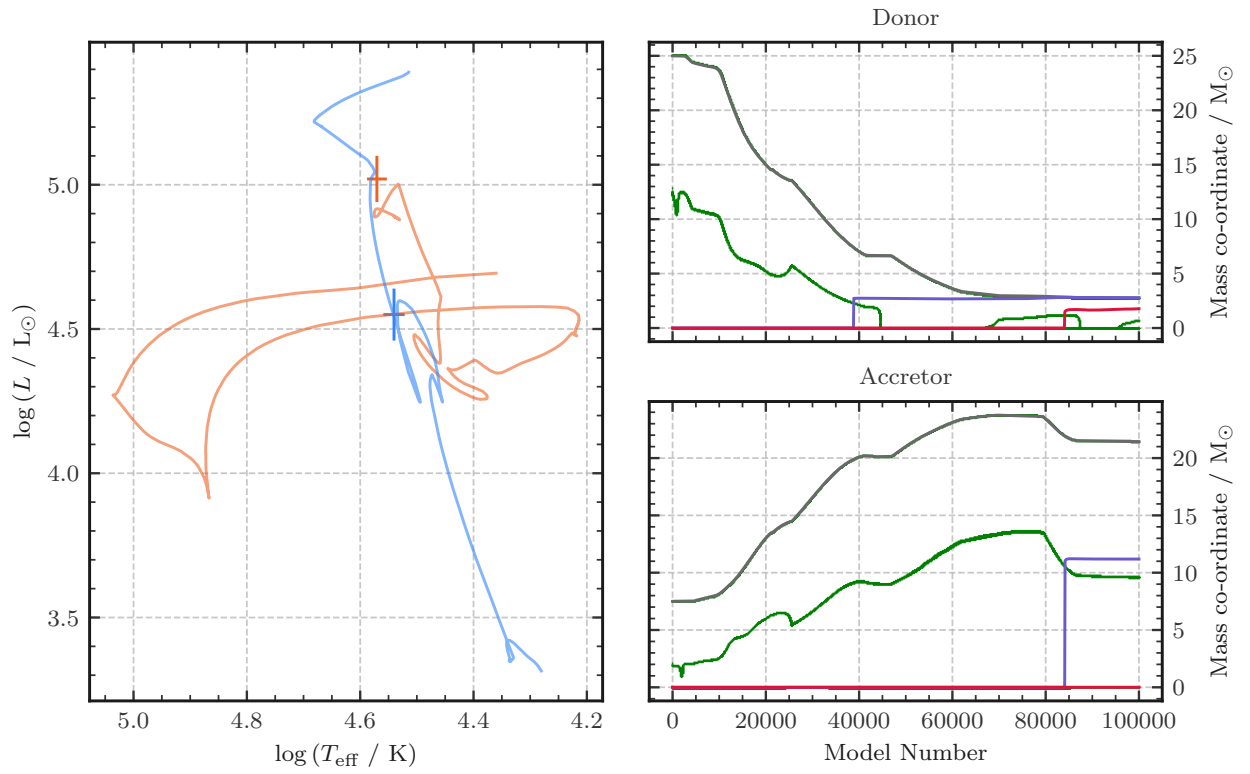
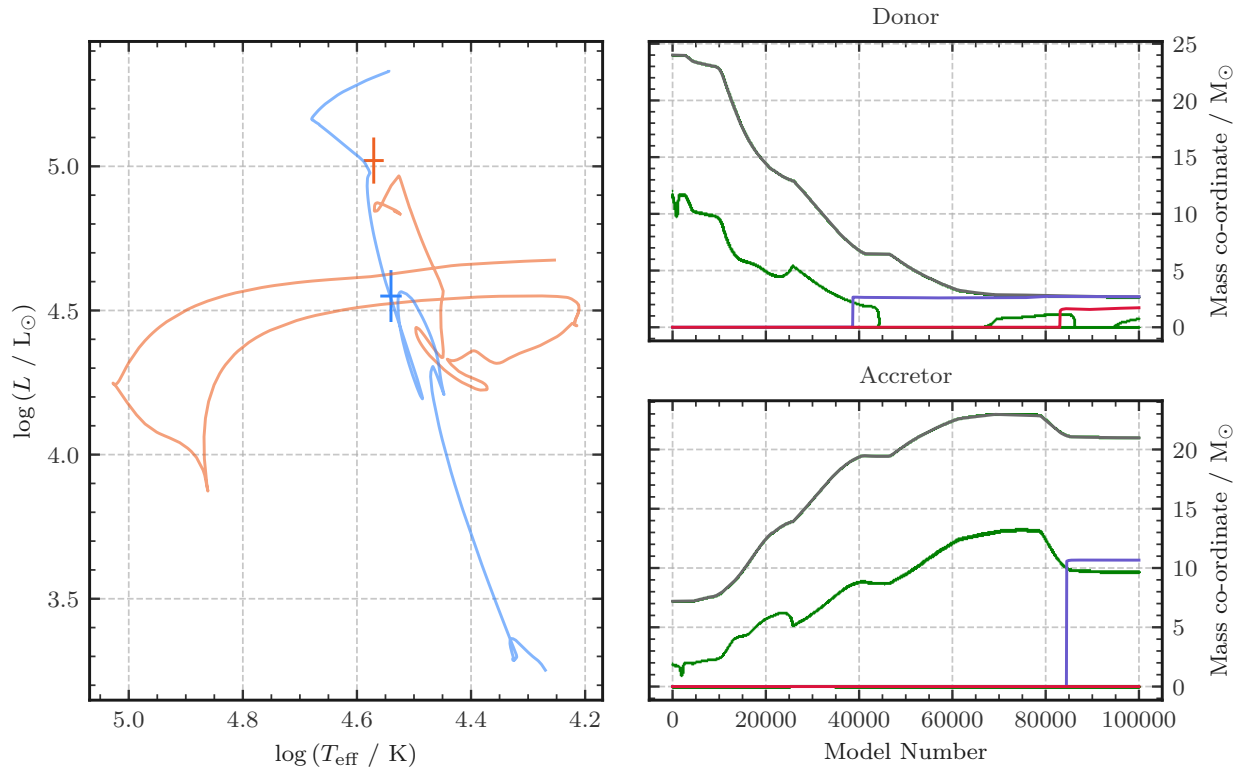


Figure 5.5: HRDs for the matching overcontact binary models. Also shown are the crosses representing the locations of the observed systems as reported in their original papers. The secondary luminosities in particular are not well constrained, and thus their locations must be taken with caution.



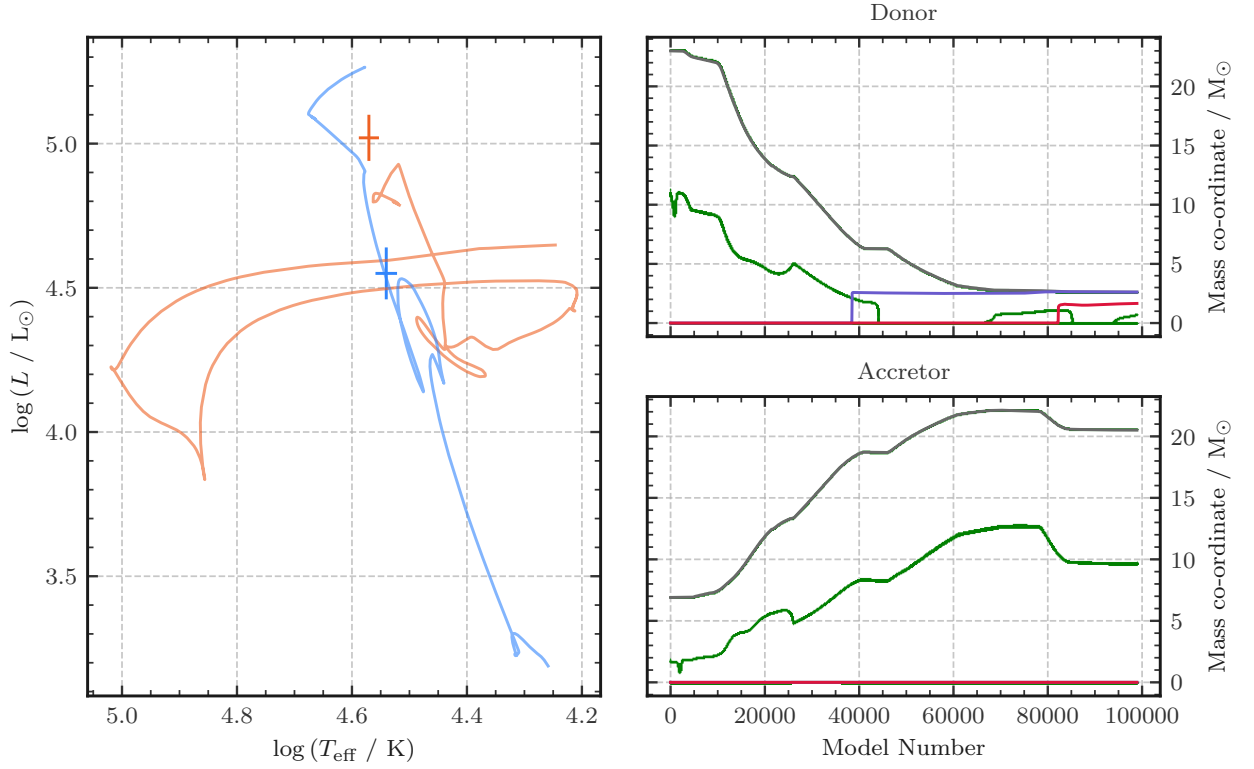
(a) The first match for TU Muscae ($M_1 = 25 M_{\odot}$, $M_2 = 7.5 M_{\odot}$, $\log(P / d) = 0.3$).



(b) The second match for TU Muscae ($M_1 = 24 M_{\odot}$, $M_2 = 7.2 M_{\odot}$, $\log(P / d) = 0.3$).

Binary	ZAMS parameters			Observed details						Match details			Matches	
	$M_1 / M_\odot$	q	$\log P / \text{d}$	$M_1 / M_\odot$	$R_1 / R_\odot$	$M_2 / M_\odot$	$R_2 / R_\odot$	$M_1 / M_\odot$	$R_1 / R_\odot$	$M_2 / M_\odot$	$R_2 / R_\odot$	*1	*2	
LSS 3074	30	0.3	0.7	14.8 ± 2.2	7.5 ± 0.6	17.2 ± 4.2	8.2 ± 0.7	13.37 ± 0.01	7.40 ± 0.33	17.96 ± 0.00	4.82 ± 0.00	✓	✕ _R	
LSS 3074	30	0.3	0.5	14.8 ± 2.2	7.5 ± 0.6	17.2 ± 4.2	8.2 ± 0.7	13.73 ± 0.63	8.03 ± 0.03	18.39 ± 0.27	4.82 ± 0.03	✓	✕ _R	
TU Mus	25	0.3	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	16.50 ± 0.59	7.20 ± 0.29	11.55 ± 0.54	6.17 ± 0.06	✓	✓	
TU Mus	24	0.3	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	16.01 ± 0.57	7.24 ± 0.27	10.92 ± 0.50	6.15 ± 0.04	✓	✓	
TU Mus	23	0.3	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.77 ± 0.41	7.42 ± 0.17	10.07 ± 0.32	6.13 ± 0.01	✓	✓	
TU Mus	16	0.6	0.5	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.91 ± 0.01	7.18 ± 0.29	9.58 ± 0.00	4.36 ± 0.04	✓	✕ _R	
TU Mus*	16	0.6	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.90 ± 0.01	7.18 ± 0.29	9.58 ± 0.00	4.37 ± 0.04	✓	✕ _R	
TU Mus*	16	0.6	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.12 ± 0.01	7.69 ± 0.01	10.33 ± 0.01	5.02 ± 0.01	✓	✕ _R	
V382 Cyg	50	0.2	0.7	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	27.18 ± 0.19	9.43 ± 0.11	21.16 ± 0.17	4.79 ± 0.04	✓	✕ _R	
V382 Cyg	40	0.3	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	24.70 ± 0.19	9.40 ± 0.11	20.05 ± 0.19	6.40 ± 0.30	✓	✕ _R	
V382 Cyg	40	0.4	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	23.61 ± 0.41	9.54 ± 0.03	26.69 ± 0.21	6.72 ± 0.02	✓	✕ _{MIR}	
V382 Cyg	30	0.4	0.7	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.24 ± 0.03	9.43 ± 0.10	12.00 ± 0.00	4.70 ± 0.01	✓	✕ _{MIR}	
V382 Cyg	30	0.4	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.17 ± 0.04	9.40 ± 0.12	12.00 ± 0.00	4.70 ± 0.01	✓	✕ _{MIR}	
V382 Cyg	30	0.5	0.7	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.24 ± 0.03	9.42 ± 0.10	14.98 ± 0.00	5.48 ± 0.02	✓	✕ _R	
V382 Cyg	30	0.5	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.17 ± 0.04	9.40 ± 0.12	14.98 ± 0.00	5.49 ± 0.02	✓	✕ _R	

Table 5.6: The observed vs. matched parameters for each of the 14 overcontact matches we find. An asterisk next to the star name indicates that the binary undergoes two periods where it matches the donor mass / radius, and each match is shown. In the “matches” column, ✓ represents that both the masses and the radii overlap for a given model, and ✗ represents that at least one of the parameters does not match. In these instances, the parameter(s) which do not match are shown as subscripts next to the cross.



(c) The third match for TU Muscae ($M_1 = 23 M_\odot$, $M_2 = 6.9 M_\odot$, $\log(P/d) = 0.3$).

Figure 5.6: The three fits for TU Muscae we recover. Shown here are the HRDs and the Kippenhahn diagrams for both the donor and the accretor. In the Kippenhahn diagrams, the blue line is the helium core, and the red is the CO core.

5.3 | Matches to observed close contact models

The matches that we find are shown in Table 5.6 next to their search parameters. Three of the binaries have matches on mass and radius for both stars simultaneously. The balance generally do not match on the companion radius, with three systems also failing to match on the companion mass as well. We consider a mass or radius to have overlapped to its observed value if the error bars at all overlap, and caveat that the reported error bars on the observed parameters are the search uncertainties, as per Table 5.1.

The three systems which match on both the primary and secondary mass and radius all model TU Muscae. We show in Figure 5.6 the HRDs and Kippenhahn diagrams of these three matches. The relative positions of the observations on the HRD suggest that a mass ratio inversion occurred for this system, and thus our labellings of the primary / secondary are incorrect.

The three matches for TU Muscae differ only by their initial masses. The primary masses are $25 M_\odot$ (Figure 5.6a), $24 M_\odot$ (Figure 5.6b), and $23 M_\odot$ (Figure 5.6c). All three have initial periods of $10^{0.3} \text{ d} \sim 2 \text{ d}$, and all three have mass ratios of $1/3$. Of these three matches, the closest match is the $25 M_\odot$ progenitor in Figure 5.6a.

What then remains is to evaluate the goodness-of-fit of the detailed model code results. The three results are summarised in Table 5.7. The three models all fulfil their matching criteria at similar ages to their v2 counterparts. The differences can be ascribed to two key factors. Firstly, the BPASS v2 ages were computed

as matches on the HR diagram to primarily the donor, as BPASS v2 models the secondary as a point mass during the pre-supernova evolution of the donor. This is in contrast to the STARS matches, which matched masses and radii of *both* stars. Secondly, the ZAMS periods were 0.1 dex wider than the ones used in the computation of the STARS model, and originate in our lower-period grid as described at the beginning of Section 5.2.

Binary	ZAMS Parameters			BPASS v2		STARS code	
	$M_1 / M_\odot$	q	$\log P / \text{d}$	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$
TU Mus	25	0.3	0.3	6.69	0.56	6.5174	0.0002
TU Mus	24	0.3	0.3	6.71	0.56	6.5509	0.0002
TU Mus	23	0.3	0.3	6.77	0.52	6.5843	0.0001

Table 5.7: The match results with associated mean ages at match. The BPASS v2 periods were computed with 0.1 dex periods higher, hence a slight mismatch in the ages is not unexpected. Ages from BPASS v2 were first presented in Table 5.2, and ages from STARS are computed in the same way, although on masses and radii compared to temperatures and luminosities.

5.3.1 | The length of the contact phase

We also briefly investigate the mean length of the contact phase. For each binary in our sample, we determine the age at which it fills both Roche lobes, and the age at which it detaches. If the binary undergoes two contact phases, as is the case in one model of TU Muscae (see Table 5.5), we record this information for each phase. If the binary is still in a contact phase at the end of the model sequence, we record the final age as the termination age. We find that as the initial period increases, the mean contact time of the binary decreases, as expected. We find that the mean contact time in the widest period grid is $\mu_{\text{wide}} = 3.819 \text{ kyr}$, $\sigma_{\text{wide}} = 1.723 \text{ kyr}$, of the normal period grid $\mu_{\text{normal}} = 5.871 \text{ kyr}$, $\sigma_{\text{normal}} = 2.773 \text{ kyr}$, and of the shorter period grid $\mu_{\text{wide}} = 20.897 \text{ kyr}$, $\sigma_{\text{wide}} = 73.755 \text{ kyr}$. We show in Figure 5.7 the dependence of the mean contact length on the initial parameters. For (comparatively) low initial primary masses $M_1 \lesssim 24 M_\odot$, there is no dependence. Comparatively, we can see that the mean contact length generally decreases with initial period and mass ratio.

5.4 | Discussion

The most obvious limitation is that we were able to only produce three matching systems on one of the six close-contact binaries. Figure 5.3 shows that, especially for SMC 108086 and MY Camelopardalis, the parameter space exploration was insufficient and that, were the space widened, more matches may be able to be found.

In addition, the systems here were assumed to be initially nonrotating. Adjusting the initial rotation rate of either or both stars may allow us to obtain a better fit, however this immediately increases the number of models one would need to consider, as there is no immediate prior on what the initial rotation rate of the star(s) would need to be.

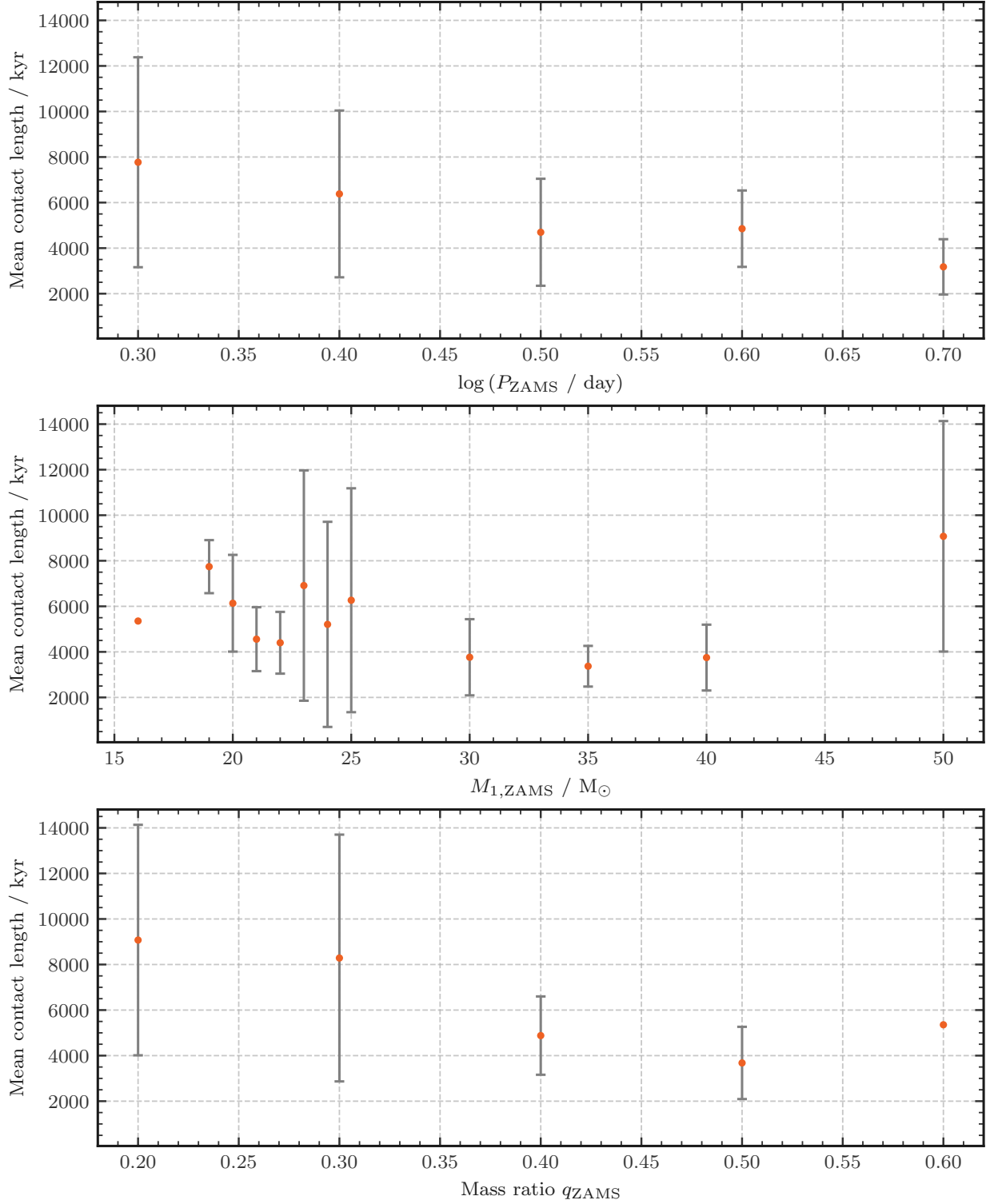


Figure 5.7: The dependence of the mean contact length on the initial parameters $\log P$, M_1 , and q . Error bars correspond to the standard deviations (i.e., 1σ uncertainties) of the measurements. The data points with no error bar only contain one model. One outlier, corresponding to $M_1 = 50 M_{\odot}$, $q = 0.2$, $\log(P / \text{d}) = 0.5$, was manually removed, as it had a contact length of ~ 400 Myr, and rendered the rest of the plot unreadable.

The initial matches for systems, as listed in Table 5.1, were performed as matches to positions on an HRD, which presents difficulty. First, as contact binaries, the secondary luminosity is difficult to constrain, and the assumption of the Stefan-Boltzmann law (Eqn(5.1)) is unlikely to hold proper. This resulted in intentionally wide search bars, to encapsulate this uncertainty, but also resulted in binaries that are unlikely to actually match the search parameters.

This study also relies on the BPASS v2 models to evaluate the ZAMS parameters of our models. BPASS uses the approximations by [Hurley et al. \(2000\)](#) to approximate the evolution of the secondary star, contributing further to the aforementioned difficulty in constraining the search parameters for the accretor.

In order to obviate these limitations, a wide grid in both primary and secondary mass could be computed. By doing so, one removes the dependence on the BPASS secondary grid and inherent uncertainties arising from [Hurley et al. \(2000\)](#), at the computational expense of having to search a much larger range of models for matches.

5.5 | Chapter Summary

We started by compiling the six contact binaries in the sample of [Abdul-Masih et al. \(2022\)](#). We matched these binaries to their ZAMS parameters using the BPASS v2.2 and HOKI codes, finding between one and ten viable sets of initial parameters to each model. We also found their mean ages to be between 1.34 Myr and 13.8 Myr, depending on the model. We also used the BPASS v2.2 models to predict the eventual outcome of these contact binaries, and found that a range of possible compact object binaries could occur, with initial periods on the order of tens of years.

By using the BPASS fits as a starting point for the initial parameters of the contact binaries, we created bespoke binary evolution models for the contact binaries with the STARS code, we found that 14 model tracks matched, for LSS 3074, TU Muscae, and V382 Cygni. We found that these matches were short compared to the lifetime of the binary, generally existing for ~ 200 kyr – ~ 600 kyr. Of the 14 matches, we found that only three matched the observed masses and radii of both components in the binary, all for TU Muscae on the shortest grid. The models for LSS 3074 and V382 Cygni, as well as three of the models for TU Muscae were unable to match the radius of the secondary, and three of the models for V382 Cygni were unable to match the secondary mass as well. We conjectured that this may be an artifact of the parameter space explored, and that a wider grid would enable us to find closer matches.